

Unveiling *Super Experts* in Mixture-of-Experts Large Language Models

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Abstract

Sparsely activated Mixture-of-Experts (MoE) models have shown promise in enhancing the learning capacity of large language models (LLMs), yet their substantial number of parameters presents significant challenges for the deployment of MoE LLMs. Leveraging the intrinsic importance differences among experts, recent research has explored expert-level compression techniques to improve the efficiency of MoE LLMs. However, existing approaches often rely on empirical criteria to identify critical experts, lacking a deeper exploration and understanding of the heterogeneous importance of experts. In this study, we present the first discovery and investigation of a distinct subset of experts that play a crucial role in the underlying mechanisms during the model’s forward inference. These experts are prevalent in open-source MoE LLMs, and despite their limited number, pruning them leads to a significant decline in model performance (e.g., pruning three causes Qwen3-30B-A3B to produce repetitive and uninformative outputs). We refer to these experts as **Super Experts** (SEs). Our comprehensive analysis provides progressively deeper insights into SEs. (i) SEs are characterized by rare but extreme activation outliers in the output of the `down_proj`, which give rise to massive activations in the hidden states between decoder layers. Moreover, the distribution of SEs remains model-specific and is unaffected by post-training processes. (ii) By pruning SEs, we assess their significance across a variety of tasks, revealing their considerable impact on the model’s overall performance, particularly in mathematical reasoning. (iii) We further enhance our understanding of the influence of SEs compression. Our findings confirm that MoE LLMs rely on SEs to induce attention sinks, which are crucial for the distribution of attention scores but are significantly disrupted by SE pruning. We also develop an automated tool for the rapid and precise profiling of SEs in new MoE LLMs. The code is available at <https://github.com/ZunhaiSu/Super-Experts-Profilling>.

1 Introduction

Sparsely activated Mixture-of-Experts (MoE) models employ dynamic routing and sparse activation, demonstrating significant potential in enhancing the learning capacity of large language models (LLMs) (Cai et al. 2024; Mu and Lin

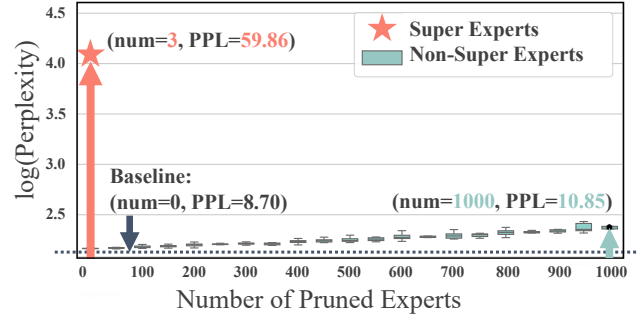


Figure 1: Analysis of experts pruning on Qwen3-30B-A3B using the WikiText-2 dataset. Pruning three Super Experts results in a significant degradation in perplexity (PPL).

2025). This paradigm has led to the development of state-of-the-art MoE LLMs, including DeepSeek (Guo et al. 2025; Liu et al. 2024b), Qwen (Team 2024b; Yang et al. 2025), and others (Meta 2025; Jiang et al. 2024). Despite their potential, a significant challenge with MoE LLMs stems from their large parameter size and high computational cost (Li et al. 2023; Lu et al. 2024; Chowdhury et al. 2024), which present considerable obstacles for deployment. Model compression techniques, such as quantization (Frantar et al. 2022; Xiao et al. 2023a; Su et al. 2025) and pruning (Frantar and Alishtarh 2023; Sun et al. 2023; Ma, Fang, and Wang 2023), enable the development of more compact and computationally efficient models (Zhu et al. 2024; Wang et al. 2024).

Beyond LLM-oriented compression approaches, expert-level compression methods have been developed by leveraging the structural characteristics of MoE models and the uneven importance of experts induced by training strategies (Chowdhury et al. 2024; Chi et al. 2022; Lu et al. 2024). Specifically, expert-level compression employs various expert importance metrics to guide the pruning, merging, or skipping of less critical experts (Lu et al. 2024; Huang et al. 2025; Xie et al. 2024), prioritize more important ones by assigning higher bit budgets during quantization (Duanmu et al. 2025; Li et al. 2024; Zheng et al. 2025), and allocate more ranks in low-rank decomposition (Yang et al. 2024; Li et al. 2023). For instance, several studies evaluate expert importance based on access frequency or other statistical features derived from the routers in MoE layers (Chen et al.

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2022; Li et al. 2024, 2023; Huang et al. 2025).

Analyzing expert importance not only facilitates model compression but also provides deeper insights into the inner workings of MoE LLMs. However, existing approaches often rely on empirical criteria to identify critical experts, lacking a deeper exploration and understanding of the heterogeneous importance among experts. In this study, we address a fundamental yet previously overlooked question: *Is there a distinct subset of experts that plays a critical role in the underlying mechanisms during the forward inference of MoE LLMs?* Through comprehensive analysis of various open-source MoE LLMs, we consistently confirm the existence of such experts. Despite their limited number, pruning these experts leads to a significant collapse in model performance. As shown in Figure 1, pruning just three experts from Qwen3-30B-A3B (Yang et al. 2025) leads to a significant degradation in model performance, while randomly pruning other experts results in a considerably smaller impact. We refer to these experts as **Super Experts** (SEs), and our subsequent comprehensive analysis offers increasingly deeper insights into SEs.

In Section 3, we first characterize SEs and explore their distribution across different models and input data domains. SEs are identified by rare but extreme activation outliers in the output of the `down_proj`, which induce massive activations (Sun et al. 2024) in the hidden states between decoder layers. Intriguingly, the distribution of SEs remains model-specific, and the SEs in the base model maintain consistency after post-training processes. For instance, Qwen3-30B-A3B-Base and Qwen3-30B-A3B exhibit identical SEs distribution patterns. In Section 4, we assess the importance of SEs by quantifying performance degradation across multiple capabilities following their pruning. For non-reasoning models, all few-shot evaluations we conducted demonstrate considerable performance degradation, with the most pronounced decline observed in the GSM8K dataset (Cobbe et al. 2021). Notably, for reasoning models, pruning SEs leads to a complete performance collapse, with Pass@1 dropping to nearly zero on tasks such as AIME and Math-500 (AIME 2024, 2025; Lightman et al. 2023). This highlights the critical role of SEs in the model’s overall performance, particularly in mathematical reasoning tasks.

In Section 5, we further enhance our understanding of the influence of SE compression. Our findings confirm that MoE LLMs rely on SEs to induce attention sinks, which are crucial for the distribution of attention scores and must be preserved during sparse attention or token compression (Xiao et al. 2023b; Su et al. 2025). We quantitatively assess the significant disruption of attention sinks caused by SE compression, as measured by the proposed **attention sink decay rate**. Our contributions are summarized as follows:

- We present the first discovery and investigation of **Super Experts** (SEs) in MoE LLMs. We identify and localize SEs across various prominent open-source MoE LLMs, thoroughly validating that their distribution remains model-specific and unaffected by post-training processes. Additionally, we develop an automated tool for rapid and precise profiling of SEs in new models.

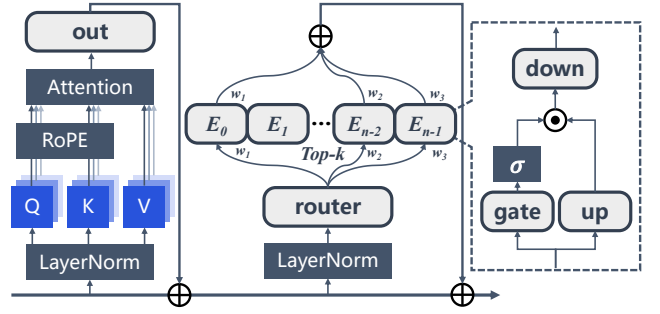


Figure 2: Decoder Architecture of MoE LLM.

- We conduct extensive experiments to evaluate the importance of SEs, revealing their significant impact on the model’s overall performance, particularly in mathematical reasoning abilities.
- We further enhance the understanding of SEs compression. Our findings confirm that MoE LLMs rely on SEs to induce attention sinks. Using the proposed **attention sink decay rate**, we quantitatively assess the significant disruption of attention sinks caused by SE pruning.
- This discovery of SEs provides new insights into the inner workings of MoE LLMs and highlights the critical importance of preserving SEs during model compression. These insights offer valuable guidance for achieving more effective MoE LLM compression strategies.

2 Preliminaries on MoE LLMs

MoE LLMs LLMs are typically structured as a stack of Transformer decoder blocks (Vaswani et al. 2017), each consisting of a multi-head self-attention (MHSA) layer and a feed-forward network (FFN) layer. In MoE LLMs, the FFN layers are replaced by MoE layers, where each layer consists of multiple experts, each represented by a FFN. A concise overview of MoE LLMs is presented in Figure 2. Let $H^0 \in \mathbb{R}^{n \times d}$ represent the input to the first decoder, where d is the embedding dimension, and n is the length of the tokenized input sequence. Then, the output of the l -th decoder block, $H^l \in \mathbb{R}^{n \times d}$, is given by:

$$H^l = \text{MoE} \left(\text{LN}_{\text{moe}} \left(H^{l'} \right) \right) + H^{l'}, \quad (1)$$

$$H^{l'} = O^l + H^{l-1}, O^l = \text{MHSA} \left(\text{LN}_{\text{mhsa}} \left(H^{l-1} \right) \right), \quad (2)$$

where $1 \leq l \leq L$, with L denoting the total number of blocks. LN refers to layer normalization, O^l representing the output of the MHSA, and $H^{l'}$ denoting residual summations after the MHSA.

MHSA Layer After LN, the input H^{l-1} is projected through the weight matrices $W_Q^l, W_K^l, W_V^l \in \mathbb{R}^{d \times d}$ to generate the Query, Key, and Value, which are then divided into K heads, denoted as $Q^{l,k}, K^{l,k}, V^{l,k}$, for $1 \leq k \leq K$. The MHSA is computed as:

$$A^{l,k} = \text{Softmax} \left(\frac{Q^{l,k} K^{l,kT}}{\sqrt{d_k}} + M \right), \quad (3)$$

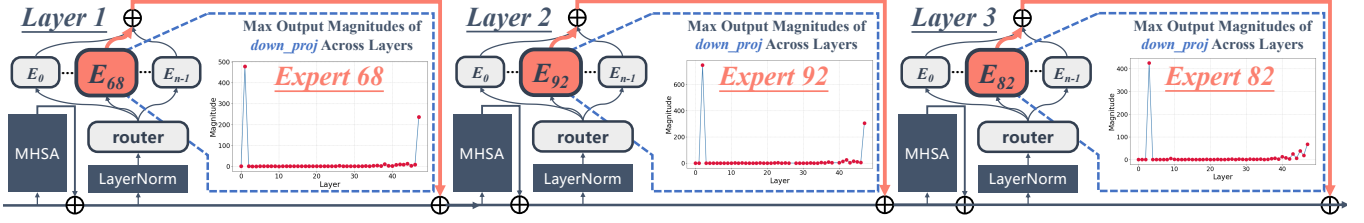


Figure 3: Super Experts mechanism in Qwen3-30B-A3B. The line plots depict the maximum output magnitudes of `down_proj` for experts 68/92/82 across layers. Massive activation is gradually amplified through expert 68 in layer 1, expert 92 in layer 2, and expert 82 in layer 3. Extreme activation outliers from these Super Experts are propagated into the hidden states between decoders via residual summation, progressively leading to massive activation.

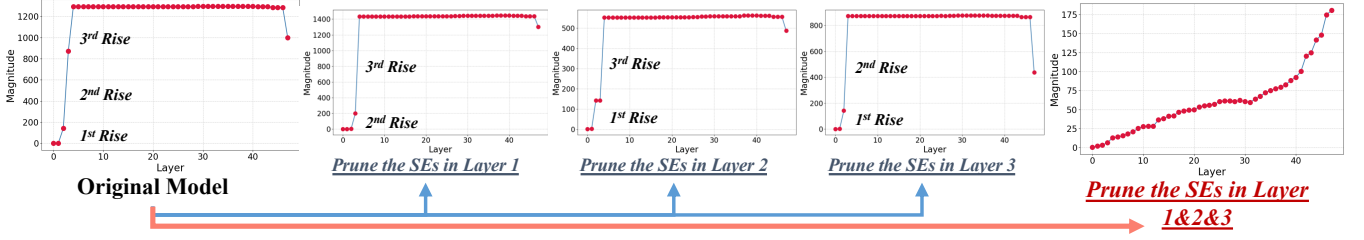


Figure 4: Impact of SEs pruning on massive activations in Qwen3-30B-A3B. Massive activations are computed using 100 input samples from the C4 (Raffel et al. 2020) dataset, each with a length of 2K.

$$O^l = \text{Concat}_{k=1}^K (A^{l,k} V^{l,k}) W_O^l, \quad (4)$$

where M represents the attention mask, and $d_k = d/K$. For simplicity, the rotation position encoding (RoPE) (Su et al. 2024) is omitted here.

MoE Layer Next, $H^{l'}$ is passed through a new LN and enters the MoE layer. First, the router network determines which experts to activate and how to scale their outputs through the weight matrix W_G . The routing weights $G \in \mathbb{R}^{n \times E}$ are computed as:

$$G = \text{softmax}(H^{l'} W_G). \quad (5)$$

Then, sparse activation of the experts is achieved by selecting the top- k routing weights for each input token. The output of the activated experts is scaled by the routing weights and aggregated to form the output of the MoE layer:

$$\sum_{i \in \text{Top-}k(G_j)} G_{ji} \cdot \text{FFN}(\text{LN}_{moel}(H_j^{l'})), \quad \forall j = 1 \dots n, \quad (6)$$

where $\text{Top-}k(G_j)$ denotes the indices of the top- k routing weights for the j -th input token. The FFN is defined as:

$$\text{FFN}(X) = (\sigma(XW_g) \odot XW_u) W_d, \quad (7)$$

where W_g , W_u , and W_d are the weight matrices for the gating, up-projection, and down-projection, respectively. σ denotes the activation function, and $\odot$ represents the Hadamard product.

3 Super Experts: Discovery and Localization

In this section, we first demonstrate the discovery process of SEs using Qwen’s latest MoE LLM, Qwen3-30B-A3B, as an example. Next, we analyze SEs across different MoE LLMs and data domains to examine their distribution patterns and highlight the widespread presence of SEs.

3.1 Exploring Super Experts

Massive Activations in MoE LLMs The discovery of SEs arises from an exploration and analysis of the formation of massive activations (Sun et al. 2024) in MoE LLMs. Recent research has explored a distinct class of extreme activation outliers in LLMs, which appear in the hidden states between decoder layers and are known as massive activations (Sun et al. 2024; Guo et al. 2024; Yu et al. 2024). They are limited in number, yet their values are orders of magnitude larger than those of other activations (e.g., up to 100,000 times larger). Existing research has yet to extensively explore massive activations in MoE LLMs and the mechanisms behind their formation. To address this gap, we first validate the existence of massive activations across several open-source MoE LLMs. Our findings reveal that massive activations persist in all MoE LLMs we investigated, as shown in the first image on the left in Figure 4.

Super Experts Induce Massive Activations Given that massive activations are also present in MoE LLMs, a deeper question arises: *How are these massive activations formulated in MoE LLMs?* Are they the result of all activated experts working together, or are they driven by only a few specific experts, or perhaps caused by other parts of the model? Through observations of several prominent open-source MoE LLMs (e.g., Qwen series, DeepSeek series, Mistral), we surprisingly find that a small subset of experts consistently generates rare yet extreme activation outliers in the output of their `down_proj` layers. These outliers are subsequently passed onto the hidden states via residual summation after the MoE layers, leading to massive activations. The entire process is illustrated in Figure 3 using the Qwen3-30B-A3B example. This phenomenon typ-

Model	Total Experts	SE Count	SE Proportion	Top 1	Top 2	Top 3	Top 4	Top 5	Top 10	Top 0.5%	Top 1 * 0.1
Qwen3-30B-3B	6144	3	0.05%	744.0	540.0	430.0	63.5	19.1	12.1	7.3	74.4
DeepSeek-R1	15677	10	0.06%	616.0	536.0	171.0	143.0	143.0	67.0	36.75	61.6
DeepSeek-V2-Lite-Chat	1782	2	0.22%	1424.0	462.0	112.5	89.5	37.5	24.0	34.5	142.4
Mixtral-8x7B-Instruct-v0.1	256	1	0.39%	5600.0	302.0	286.0	258.0	253.0	139.0	5600.0	560

Table 1: The activations identified as Super Experts are highlighted in bold.

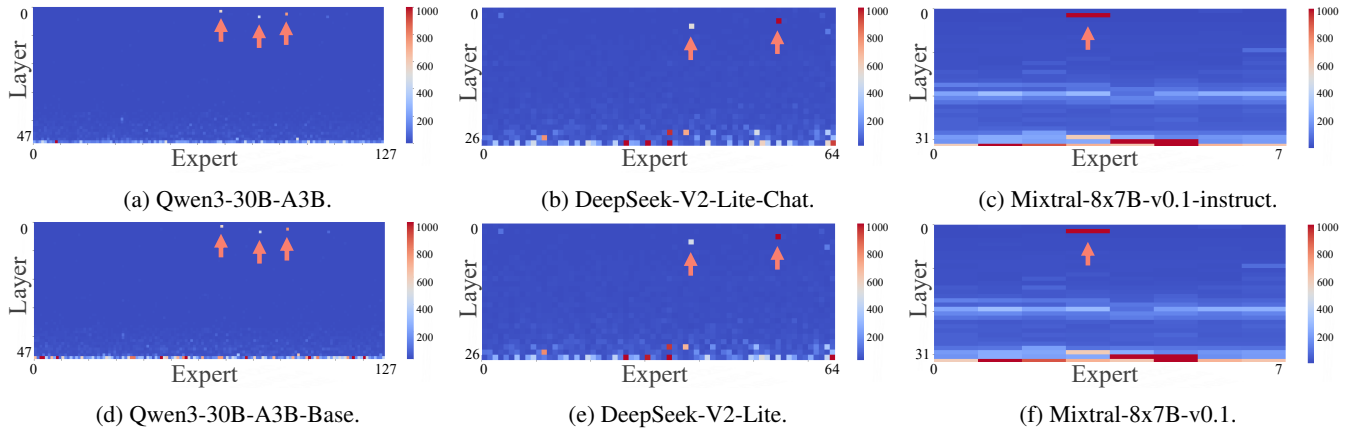


Figure 5: Heatmap visualizations of the maximum output magnitudes from the `down_proj` for each expert across layers. The analysis is conducted using the C4 (Raffel et al. 2020) dataset. SEs are highlighted with arrows.

ically occurs in a single layer (e.g., Mixtral) or in just a few layers (e.g., Qwen3-30B-A3B) starting from the initial decoder layers, ultimately leading to stable massive activations across nearly all subsequent layers. To directly validate this mechanism, we also perform ablation experiments by pruning the SEs in Qwen3-30B-A3B. As illustrated in Figure 4, pruning SEs from a single layer effectively eliminates their contribution to massive activations. Furthermore, when all SEs are pruned, massive activations are completely eliminated, confirming that they are directly generated by SEs.

3.2 Localization of Super Experts

In this section, we begin by discussing how to accurately identify SEs and then examine their distribution patterns across various models and data domains.

Super Experts Profiling Given that SEs are defined by their influence on the formation of massive activations through the extreme activation outliers they generate, we propose the following broad yet effective quantitative definition. Specifically, we compute the maximum output magnitudes to the `down_proj` for all experts across all layers. Let L denote the set of layers responsible for the formation of massive activations. Let $a_{l,e}$ denote the maximum output magnitude to the `down_proj` of expert e in layer l , and let $\mathcal{A} = \{a_{l,e}\}$ be the set of all such values across the entire model. An expert e in layer l is classified as a SE if:

$$a_{l,e} > P_{99.5} \quad \text{and} \quad a_{l,e} > \frac{1}{10} a_{\max} \quad \text{and} \quad l \in L \quad (8)$$

where $P_{99.5} = \text{Percentile}_{99.5}(\mathcal{A})$ and $a_{\max} = \max \mathcal{A}$. This criterion effectively identifies the experts of interest across

Model	Super Experts
Qwen3-30B-A3B	Layer 1 Expert 68, Layer 2 Expert 92,
Qwen3-30B-A3B-Base	Layer 3 Expert 82
DeepSeek-V2-Lite-Chat	Layer 2 Expert 54, Layer 3 Expert 38
DeepSeek-V2-Lite	
Mixtral-8x7B-Instruct-v0.1	Layer 1 Expert 3
Mixtral-8x7B-v0.1	

Table 2: Distributions of SEs in Several MoE LLMs.

various MoE LLMs we investigated, as highlighted in bold in Table 1. We have developed an automated tool for rapid and precise SE profiling based on this definition. The code is available at <https://github.com/ZunhaiSu/Super-Experts-Profiling>.

Distribution of SEs Across Different Models We select three representative MoE LLMs with distinct designs for analysis: Qwen3-30B-A3B, DeepSeek-V2-Lite-Chat, and Mixtral-8x7B-Instruct-v0.1. We also include the base models of these three LLMs to illustrate the impact of post-training processes. Although all of these models are MoE LLMs, they exhibit distinct design differences. For instance, Qwen3 and Mixtral do not employ shared experts, whereas DeepSeek does. DeepSeek-V2-Lite adopts a hybrid architecture, wherein the first layer utilizes dense MLPs, while the remaining layers are based on MoE blocks. Through the proposed SE profiling tool, we identify the SEs in these models. A summary of the SEs is provided in Table 2, and heatmap visualizations of the maximum output magnitudes from the `down_proj` are shown in Figure 5. The key con-

Model	Setting	PPL	Avg.	ARC-c	ARC-e	BoolQ	GSM8K	HellaSwag	MMLU	OpenBookQA	PIQA	WinoGrande
Qwen3 30B-A3B	Baseline	8.70	70.22	52.65	79.5	88.72	89.61	59.63	77.82	34.2	79.33	70.56
	Prune SEs	59.86	0.5500	46.08	76.05	70.73	42.38	39.31	56.03	29.8	72.52	62.12
	Drop Rate (%)	-	21.68%	12.48%	4.34%	20.28%	52.71%	34.08%	28.00%	12.87%	8.58%	11.96%
DeepSeek V2-Lite	Random Pruning	8.71	70.36	52.73	79.46	88.59	89.84	59.5	77.84	34	79.76	71.51
	Drop Rate (%)	-	-0.20%	-0.15%	0.05%	0.15%	-0.26%	0.22%	-0.03%	0.58%	-0.54%	-1.35%
	Baseline	6.31	60.27	46.59	78.37	79.79	37.83	58.75	55.03	34.6	80.3	71.19
Mixtral 8x7B-v0.1	Prune SEs	10.75	43.90	29.27	54.92	68.62	9.78	43.72	41.77	21	68.28	57.7
	Drop Rate (%)	-	27.17%	37.18%	29.92%	14.00%	74.15%	25.58%	24.10%	39.31%	14.97%	18.95%
	Random Pruning	6.31	60.30	46.5	78.45	80.37	37.38	58.77	55.1	34.4	80.14	71.59
	Drop Rate (%)	-	-0.05%	0.19%	-0.10%	-0.73%	1.19%	-0.03%	-0.13%	0.58%	0.20%	-0.56%
Mixtral 8x7B-v0.1	Baseline	3.84	67.84	56.57	84.26	85.02	57.32	64.89	67.83	35.6	82.48	76.56
	Prune SEs	6.23	49.38	36.01	64.44	75.66	24.34	50.6	42.47	20.6	73.12	57.22
	Drop Rate (%)	-	27.21%	36.34%	23.52%	11.01%	57.54%	22.02%	37.39%	42.13%	11.35%	25.26%
	Random Pruning	3.86	67.82	56.57	84.09	85.23	58.15	64.92	68.08	35	82.21	76.16
	Drop Rate (%)	-	0.02%	0.00%	0.20%	-0.25%	-1.45%	-0.05%	-0.37%	1.69%	0.33%	0.52%

Table 3: Evaluation of the importance of SEs in non-reasoning models. The results of random pruning are obtained by averaging the performance over five runs, each randomly pruning the same number of experts.

Model	Setting	PPL	Avg.	GPQA Diamond	Math-500	AIME 2024	AIME 2025	LiveCodeBench
				Pass@1	Pass@1	Pass@1	Pass@1	Pass@1
DeepSeek-R1	Baseline	3.33	75.64	71.50	97.60	79.33	66.33	63.44
	Prune SEs	5.18	1.81	5.05	4.00	0.00	0.00	0.00
	Drop Rate (%)	-	97.61%	93.0%	95.9%	100%	100%	100%
DeepSeek-R1	Random Pruning	3.35	75.53	72.63	98.00	77.67	67.00	62.37
	Drop Rate (%)	-	0.32%	-2.4%	-0.4%	2.1%	-1.0%	1.7%

Table 4: Evaluation of the importance of SEs in DeepSeek-R1.

clusions regarding SEs are summarized as follows:

(i) SEs are consistently present across the investigated models, accounting for less than 0.5% of all experts.

(ii) After post-training processes, the distribution of SEs remains unchanged compared to the base model.

(iii) Some experts in the final layers also exhibit extreme activation outliers. However, since they do not contribute to the formation of massive activations, they do not hold the same level of significance as SEs. Additional experimental results are available in Appendix A.

Distribution of SEs Across Different Data Domains We also examine their patterns across various input data domains. Specifically, in addition to the C4 dataset (Raffel et al. 2020), we analyze SE distributions across several other datasets, including WikiText-2 (Merity et al. 2016), C-Eval (Huang et al. 2023), GSM8K (Cobbe et al. 2021), and HumanEval (Chen et al. 2021). As shown in the Appendix B, the distribution of SEs remains highly stable, regardless of variations in the input data domain.

4 The Importance of Super Experts

In this section, we evaluate the significance of SEs by quantifying the performance degradation across various capabilities following their pruning. We use the original model and the average results from random pruning of an equivalent

number of experts as baselines. To more effectively evaluate the importance of SEs, we utilize distinct benchmark types for non-reasoning and reasoning models. For non-reasoning models, we conduct a series of few-shot evaluations, whereas for reasoning models, we employ more demanding benchmarks, encompassing tasks such as mathematical reasoning and coding.

4.1 Impact on Non-Reasoning Models

For non-reasoning models, we select three models: the non-thinking mode of Qwen3-30B-A3B, DeepSeek-V2-Lite and Mixtral-8x7B-v0.1. We utilize the datasets listed below and conduct evaluations using lm-eval (Gao et al. 2024), including ARC-challenge (ARC-c), ARC-easy (ARC-e) (Clark et al. 2018), BoolQ (Clark et al. 2019a), GSM8K (Cobbe et al. 2021), HellaSwag (Zellers et al. 2019), MMLU (Hendrycks et al. 2021), OpenBookQA (Mihaylov et al. 2018), PIQA (Bisk et al. 2020), and WinoGrande (Keisuke et al. 2019). As shown in Table 3, pruning only a few super-experts leads to significant degradation across all tasks, with average accuracy dropping by 21.68% to 27.21%. In particular, for GSM8K, the degradation ranges from 52.71% to 74.15%. In contrast, random pruning has a negligible impact, underscoring the crucial role of SEs.

Model	Setting	PPL	Avg.	GPQA Diamond	Math-500	AIME 2024	AIME 2025	HumanEval
				Pass@1	Pass@1	Pass@1	Pass@1	Pass@1
Qwen3-30B-A3B	Baseline	8.70	69.37	61.62	88.00	80.00	73.33	43.90
	Prune SEs Drop Rate (%)	59.86 -	4.02 93.62%	18.69 69.7%	1.40 98.4%	0.00 100%	0.00 100%	0.00 100%
	Random Pruning Drop Rate (%)	8.71 -	69.33 0.32%	61.62 0.00%	89.00 -1.10%	80.00 0.00%	73.33 0.00%	42.70 2.7%

Table 5: Evaluation of the importance of SEs in Qwen3-30B-A3B.

DeepSeek-R1	Problem	Answer
Original Model	How many of the first one hundred positive integers are divisible by 3, 4, and 5? Please reason step by step, and put your final answer within <code>\boxed{}</code> .	Okay, so the problem is asking: How many of the first one hundred positive integers are divisible by 3, 4, and 5? Hmm, let me think. First, I need to understand
Prune Super Experts		Okay, let's see here. So the question is how many of the first one hundred positive integers are divisible by 3, 4, and 5 right? Okay, so first, that's probably going to, like, but, that's, the way, it's, the way, it's, the way, it's, the way, it's, the way, it's, the way, it's, Repeating the way, it's, the way, it's, the way, it's, the way, it's
Random Pruning		Okay, so the problem is asking: How many of the first one hundred positive integers are divisible by 3, 4, and 5? Hmm, let me think. First, I need to understand

Table 6: Responses of DeepSeek-R1 in the MATH-500 (Lightman et al. 2023) benchmark.

4.2 Impact on Reasoning Models

For evaluating the importance of SEs in reasoning models, we select DeepSeek-R1 and the thinking mode of Qwen3-30B-A3B. We select benchmarks more suitable for testing reasoning models and conduct evaluations based on the EvalScope (Team 2024a), a comprehensive model evaluation and performance benchmarking framework meticulously developed by the ModelScope community (Team 2023). The generation configurations used in the evaluation align with the corresponding technical reports of the models. These benchmarks are:

- **General Tasks:** We use GPQA-Diamond under a 5-shot setting. GPQA (Rein et al. 2024) is a challenging dataset of multiple-choice questions authored by domain-specific multidisciplinary experts.
- **Math & Text Reasoning:** To evaluate mathematical and logical reasoning skills, we use high-level math benchmarks, including MATH-500 (Lightman et al. 2023), AIME’24 (AIME 2024), and AIME’25 (AIME 2025).
- **Agent & Coding:** To test the model’s proficiency in coding and agent-based tasks, we use LiveCodeBench (Jain et al. 2024) and HumanEval (Chen et al. 2021).

The experimental results, presented in Tables 4 and 5, show that pruning the SEs causes a significant performance degradation, while random pruning has almost no impact. The Pass@1 scores for most tasks drop to zero, highlight-

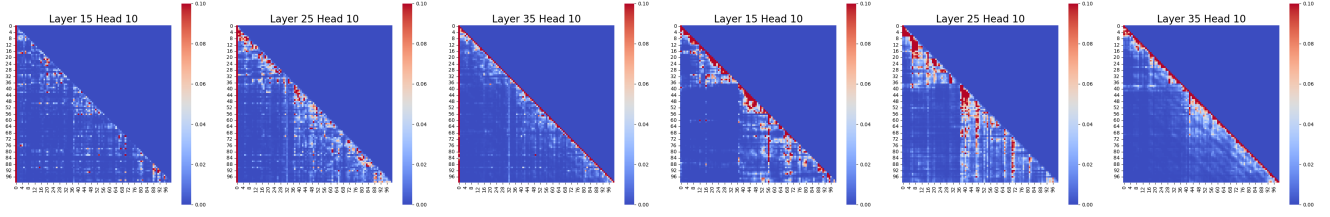
ing the critical role of SEs. During the review of model responses on the Math-500 benchmark, we made a striking observation: after pruning the SEs, the model consistently generated repetitive responses in nearly every test, continuing until it reached the maximum output length, as shown in Table 6. This behavior suggests that the model loses its ability to reason and solve problems entirely after SE pruning. More results are in Appendix C.

5 Understanding the Impact of Super Experts Compression

In this section, we focus on deepening our understanding of the underlying impact of SE compression and further quantitatively assessing its effects.

5.1 Super Experts Induce Attention Sinks

Previous studies in dense LLMs (Sun et al. 2024; Gu et al. 2024; An et al. 2025) suggest that massive activations tend to attract disproportionately high attention scores in multi-head self-attention (MHSA) layers, leading to attention sinks for initial tokens. StreamLLM (Xiao et al. 2023b) conducted an initial investigation revealing the presence of attention sinks in LLMs. Subsequent studies (Gu et al. 2024; Guo et al. 2024) have demonstrated that although attention sinks often emerge at semantically insignificant tokens, the mechanism itself is critical for model performance. In efficient LLM techniques such as sparse attention and KV cache



(a) Layer 15 Head 10. (b) Layer 25 Head 10. (c) Layer 35 Head 10. (d) Layer 15 Head 10. (e) Layer 25 Head 10. (f) Layer 35 Head 10.

Figure 6: Attention scores of Qwen3-30B-A3B. Figures (a), (b), and (c) depict the attention score maps of the original model, where the first token clearly functions as an attention sink, consistently attracting the majority of attention. Figures (d), (e), and (f) illustrate the attention scores following SE pruning, where the attention sink completely disappears.

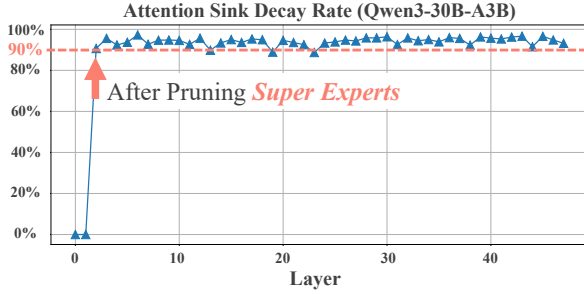


Figure 7: Attention sink decay rate across layers.

compression, maintaining attention sinks is essential for preventing undesirable distributional shifts of attention scores (Xiao et al. 2023b; Su et al. 2025). Building on this insight and our previous findings, we hypothesize that pruning SEs not only eliminates massive activations but also disrupts the underlying attention sink mechanism.

5.2 Impact of Super Experts Compression

To validate this insight and quantitatively assess the impact of SE pruning on attention sinks, we introduce *(i) Attention Sink Decay Rate*, denoted as D_{sink} . It is defined as the average decay rate of attention sinks for each head:

$$D_{\text{sink}} = 1 - \frac{1}{H} \sum_{h=1}^H \frac{\sum_{i \in S} p_i^{t'}}{\sum_{i \in S} p_i^t} \quad (9)$$

where H is the total number of heads, p_i^t represents the attention score between the Query token t and the Key token i before SE pruning, $p_i^{t'}$ denotes the attention score after SE pruning, and S refers to the set of sink tokens.

As shown in Figure 7, after pruning SEs, the attention sink decay rate remains consistently high, at approximately or even exceeding 90%, demonstrating a substantial disruptive effect on attention sinks. Figure 6 visualizes the attention scores for several heads before and after pruning SEs, highlighting the complete disappearance of attention sinks following SE pruning. Notably, attention sinks introduce implicit attention biases (Sun et al. 2024; An et al. 2025) that persist across all subsequent tokens and may encode global or other critical information (Darcet et al. 2023). Conse-

quently, the impact of SE pruning on attention computation remains both continuous and significant.

6 Related Work

6.1 Expert-Level Compression

Beyond traditional LLM-focused compression methods, recent advancements have increasingly focused on expert-level compression strategies, capitalizing on the distinct structural features of MoE LLMs and the natural variations in expert importance. These approaches—including expert pruning, skipping, merging, and expert-level mixed-precision quantization—are being actively explored for their potential to enhance MoE model efficiency. M-SMoE (Li et al. 2023) performs expert merging by using activation frequencies to consolidate less significant experts, while also applying low-rank techniques to the merged experts to achieve further compression. NAEF (Lu et al. 2024) introduces plug-and-play pruning and skipping methods that leverage reconstruction loss to selectively compress less critical experts. MC (Huang et al. 2025) harnesses the significance of both experts and tokens to perform mixed-precision quantization and dynamic expert pruning, achieving extreme compression. MC-Suite (Jaiswal et al. 2025) reviews various empirical criteria for identifying critical experts, considering four dimensions: weight, expert behavior, intermediate activations, and gradient behavior. While these methods examine expert importance from various perspectives to optimize expert compression, they lack a deeper exploration and understanding of expert significance, overlooking the mechanistic importance of specific experts. Our work presents the first discovery and investigation of *Super Experts*, offering an in-depth analysis of their mechanistic role in inference, thereby filling a critical gap in current research.

7 Conclusion

In this work, we present the first identification and systematic study of a distinct subset of experts, termed Super Experts. We analyze their characteristics, distributions, and critical functional roles within MoE LLMs. Quantitative evaluations demonstrate that compressing SEs significantly disrupts the attention sink mechanism, leading to a degradation in model performance. Future research will focus on leveraging SEs to develop more refined compression methods for MoE LLMs.

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A Further Analysis of Outlier Experts in Final Layers

Some experts in the final layers also exhibit extreme activation outliers in the output of the `down_proj`, apart from the Super Experts (SEs) in the shallower layers. We refer to these experts as outlier experts. Based on our extensive additional experiments and findings, we do not consider outlier experts to have the same mechanistic significance as SEs.

(i) We performed PPL evaluations after pruning outlier experts, and as shown in Table 7, they do not significantly affect the model’s performance in the same way as the SEs.

(ii) In both Qwen3-30B-A3B and DeepSeek-R1, pruning outlier experts does not result in repetitive outputs on reasoning benchmarks such as Math-500 (Lightman et al. 2023), whereas pruning SEs does, as shown in Tables 9 and 10. The pruned SEs and outlier experts is shown in Table 8.

(iii) We observed on Qwen3-30B-A3B that the distribution of outlier experts varies with the input dataset, while the distribution of SEs remains quite stable, as illustrated in Figure 8 and 9.

Based on our previous analysis, we infer that since massive activations occur in the shallower layers, these outlier experts are not involved in the formation of massive activations. Therefore, these experts do not operate under the same mechanism as SEs and do not hold the same level of significance. Further investigation into outlier experts is worth exploring in future work.

B Distribution of Super Experts Across Various Data Domains

In addition to analyzing the distribution of SEs across different models based on the C4 dataset (Raffel et al. 2020), we also examine their distribution patterns across various input data domains. We assess the impact of diverse language inputs on SEs using the WikiText-2 (Merity et al. 2016) and C-Eval (Huang et al. 2023) datasets. Furthermore, we investigate the influence of data from the mathematics and coding domains using the GSM8K (Hendrycks et al. 2021) and HumanEval (Chen et al. 2021) datasets. As shown in Figures 8, 9, 10, 11, 12, and 13, the distribution of SEs remains highly stable, regardless of variations in the input data domain.

C Additional Results of Reasoning Models After Super Experts Pruning

After pruning SEs, we consistently observed repetitive output and a loss of reasoning ability in both Qwen3-30B-A3B and DeepSeek-R1. The pruned SEs are shown in Table 8, and additional examples from the Math-500 (Lightman et al. 2023) benchmark are presented in Tables 9 and 10.

Model	Prune Experts	PPL	Super Experts
Qwen3-30B-A3B	Original Model	8.70	-
	Layer 1 Expert 68, Layer 2 Expert 92, Layer 3 Expert 82	59.86	Yes
	Layer 47 Expert 8, Layer 47 Expert 48, Layer 47 Expert 100	8.71	No
DeepSeek-V2-Lite	Original Model	6.31	-
	Layer 2 Expert 54, Layer 3 Expert 38	10.75	Yes
	Layer 25 Expert 11, Layer 25 Expert 39	6.32	No

Table 7: Comparison of expert pruning, with PPL evaluated using the WikiText-2 dataset.

Model	Super Experts	Outlier Experts
Qwen3-30B-A3B	Layer 1 Expert 68, Layer 2 Expert 92 Layer 3 Expert 82	Layer 1 Expert 8, Layer 47 Expert 48 Layer 47 Expert 100
DeepSeek-R1	Layer 8 Expert 24, Layer 8 Shared_expert Layer 12 Expert 190, Layer 13 Expert 64 Layer 14 Expert 202, Layer 14 Shared_expert Layer 22 Shared_expert, Layer 33 Expert 64 Layer 33 Shared_expert, Layer 35 Shared_expert	Layer 60 Expert 81, Layer 60 Expert 92 Layer 60 Expert 231, Layer 60 Shared_expert Layer 60 Expert 121, Layer 60 Expert 0 Layer 60 Expert 60, Layer 60 Expert 237 Layer 60 Expert 53, Layer 60 Expert 117

Table 8: Super Experts and Outlier Experts in Qwen3-30B-A3B and DeepSeek-R1 models.

DeepSeek-R1	Problem	Repeating	Answer
Original Model	What is the domain of the function $f(x) = \frac{2-x}{\log(2-\log(x-2))}$, where log is the base 10 logarithm function? Express your answer in interval notation. Please reason step by step, and put your final answer within <code>\boxed{\}</code> .	no	Okay, so I need to find the domain of the function $f(x) = (2 - x) / \log(2 - \log(x - 2))$. Hmm, let's see. The domain of a function includes all the real numbers x
Prune Super Experts		yes	Alright, let's try to first. the function $f(x)$ which given that we need to the domain of this function that's the which the of a that's the given this. that's the the the the the the the the the the the the the the the. that's the. that's the. that's the. that's the. that's the. Repeating that's the. that's the. that's the. that's the. that's the.
Prune Outlier Experts of Last Layer		no	Okay, let's try to figure out the domain of this function $f(x) = \frac{2-x}{\log(2-\log(x-2))}$. Hmm, domains can be tricky because there are multiple layers
Random Pruning		no	Alright, so I need to find the domain of the function $f(x) = (2 - x) / \log(2 - \log(x - 2))$. Hmm, okay. Let me start by recalling that the domain of a

Table 9: Responses of DeepSeek-R1 in the MATH-500 benchmark.

Qwen3-30B-A3B	Problem	Repeating	Answer
Original Model	Let $p(x)$ be a polynomial of degree 5 such that $p(n) = \frac{n}{n^2-1}$ for $n = 2, 3, 4, \dots, 7$. Find $p(8)$. Please reason step by step, and put your final answer within <code>\boxed{\}</code> .	no	Okay, so I need to find the value of $p(8)$ where $p(x)$ is a degree 5 polynomial that satisfies $p(n) = n / (n^2 - 1)$ for $n = 2, 3, 4, \dots, 7$
Prune Super Experts		yes	Okay, so I need to find $p(8)$ where p is a polynomial. The problem says that for each n , $p(n)$ is equal to some fraction, and we need to find $p(8)$. The question is from a competition, so it's probably from an Olympiad, and the answer is an integer. But I need to reason step by step, and put my final answer within <code>\boxed{\}</code> . Alright, let's start by understanding the problem. It says that for each n , $p(n)$ is equal to some fraction. But the problem is to find $p(8)$. The question is from a competition, so it's probably from an Olympiad, and the answer is an integer. But I need to reason step by step, and put my final answer within Repeating
Prune Outlier Experts of Last Layer		no	Okay, so I need to find the value of $p(8)$ where $p(x)$ is a degree 5 polynomial that satisfies $p(n) = n / (n^2 - 1)$ for $n = 2, 3, 4, 5, 6, 7$. Hmm, let me think about how
Random Pruning		no	Okay, so I need to find the value of $p(8)$ where $p(x)$ is a degree 5 polynomial that satisfies $p(n) = n / (n^2 - 1)$ for $n = 2, 3, 4, \dots, 7$

Table 10: Responses of Qwen3-30B-A3B in the MATH-500 benchmark after SE pruning.

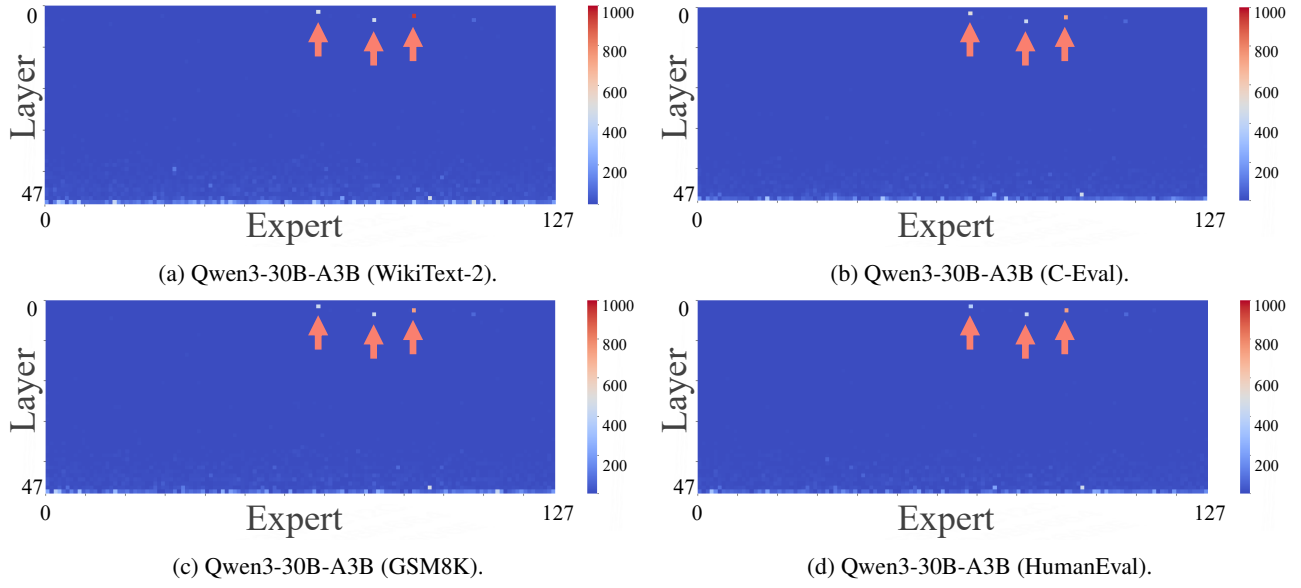


Figure 8: Heatmap visualizations of the maximum output magnitudes from the `down_proj` for each expert in Qwen3-30B-A3B across multiple datasets. SEs are highlighted with arrows.

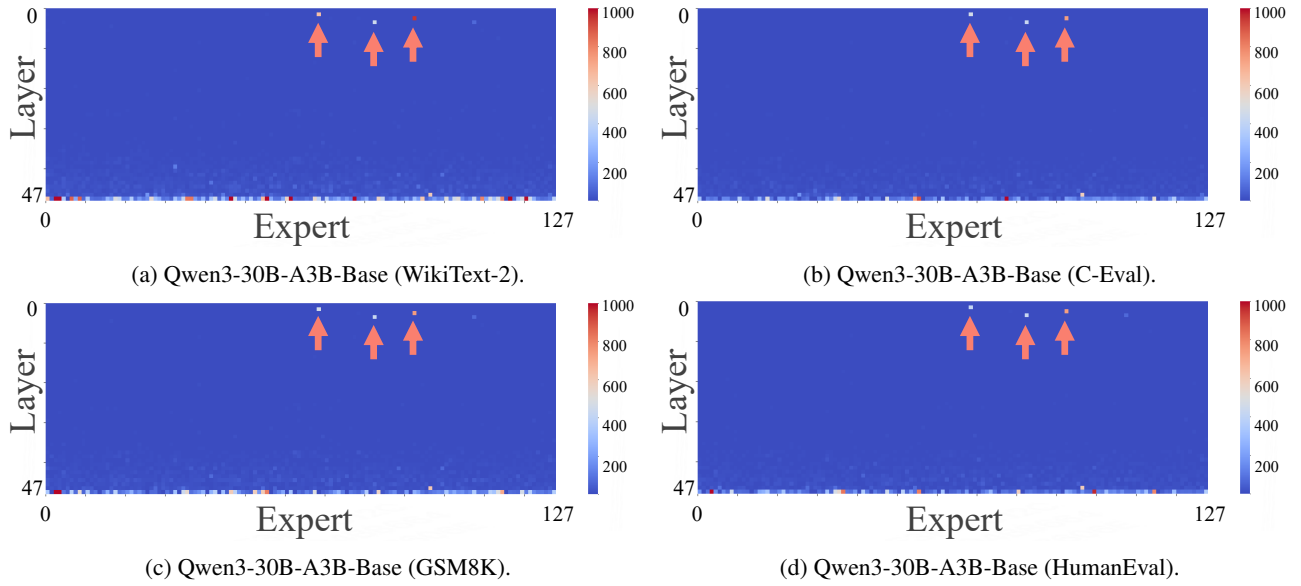


Figure 9: Heatmap visualizations of the maximum output magnitudes from the `down_proj` for each expert in Qwen3-30B-A3B-Base across multiple datasets. SEs are highlighted with arrows.

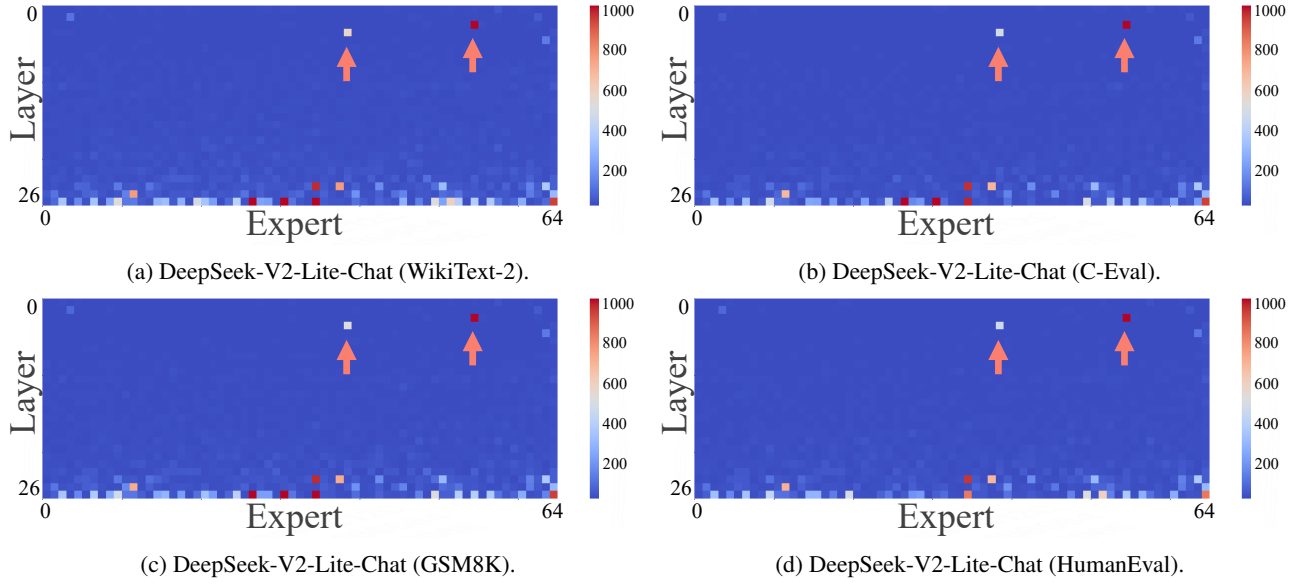


Figure 10: Heatmap visualizations of the maximum output magnitudes from the `down_proj` for each expert in DeepSeek-V2-Lite-Chat across multiple datasets. SEs are highlighted with arrows.

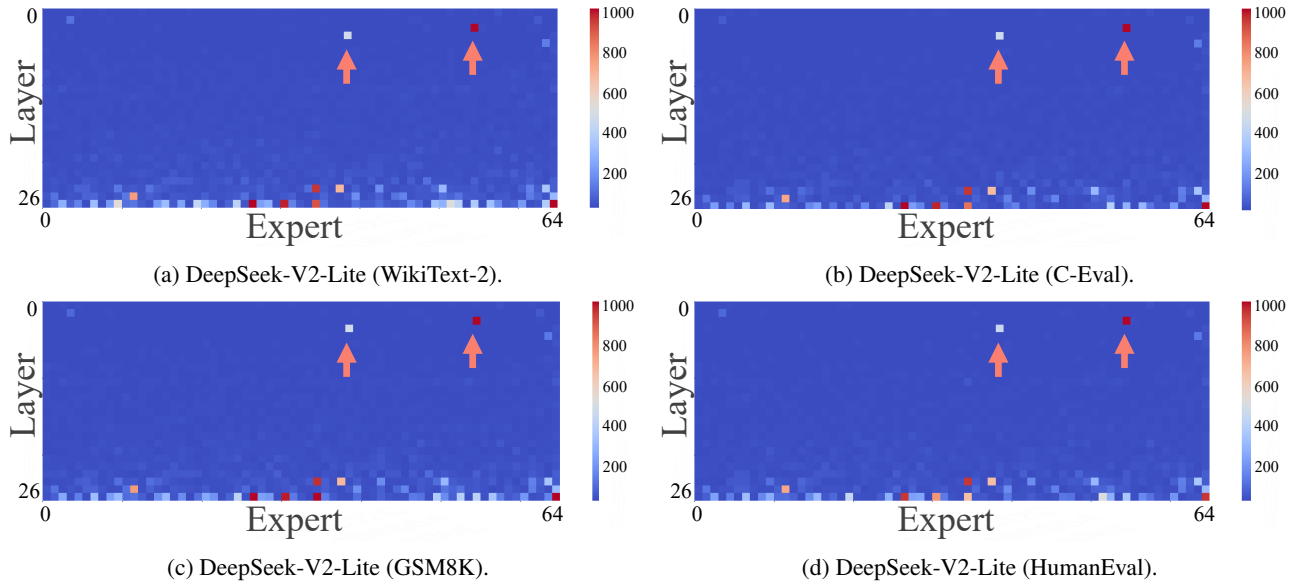


Figure 11: Heatmap visualizations of the maximum output magnitudes from the `down_proj` for each expert in DeepSeek-V2-Lite across multiple datasets. SEs are highlighted with arrows.

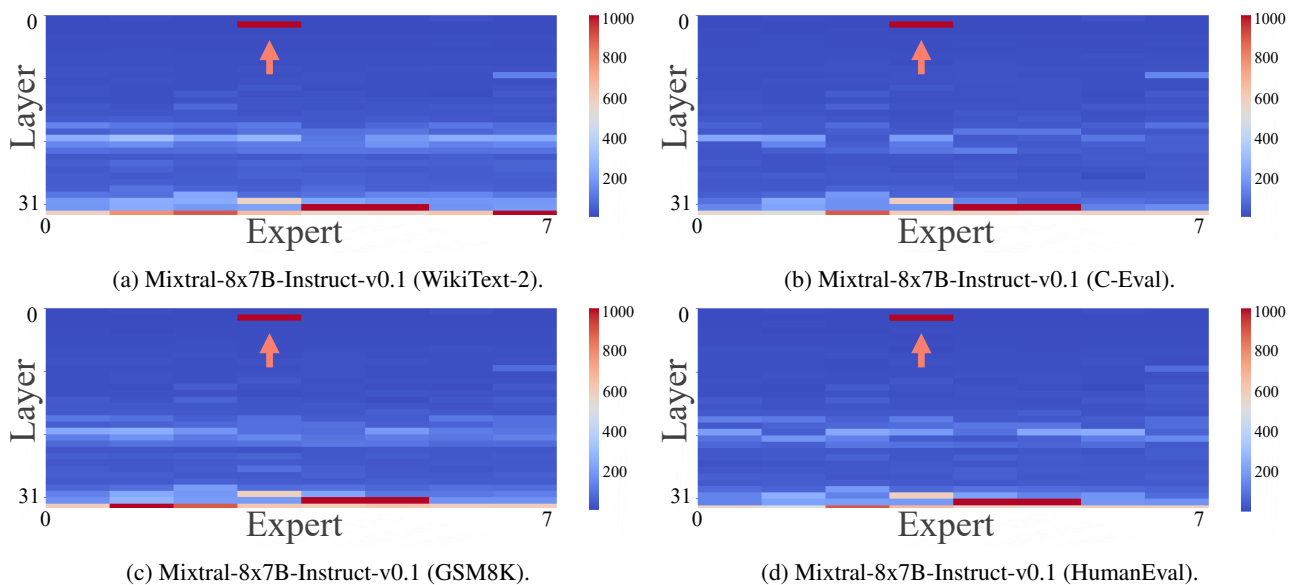


Figure 12: Heatmap visualizations of the maximum output magnitudes from the `down_proj` for each expert in Mixtral-8x7B-Instruct-v0.1 across multiple datasets. SEs are highlighted with arrows.

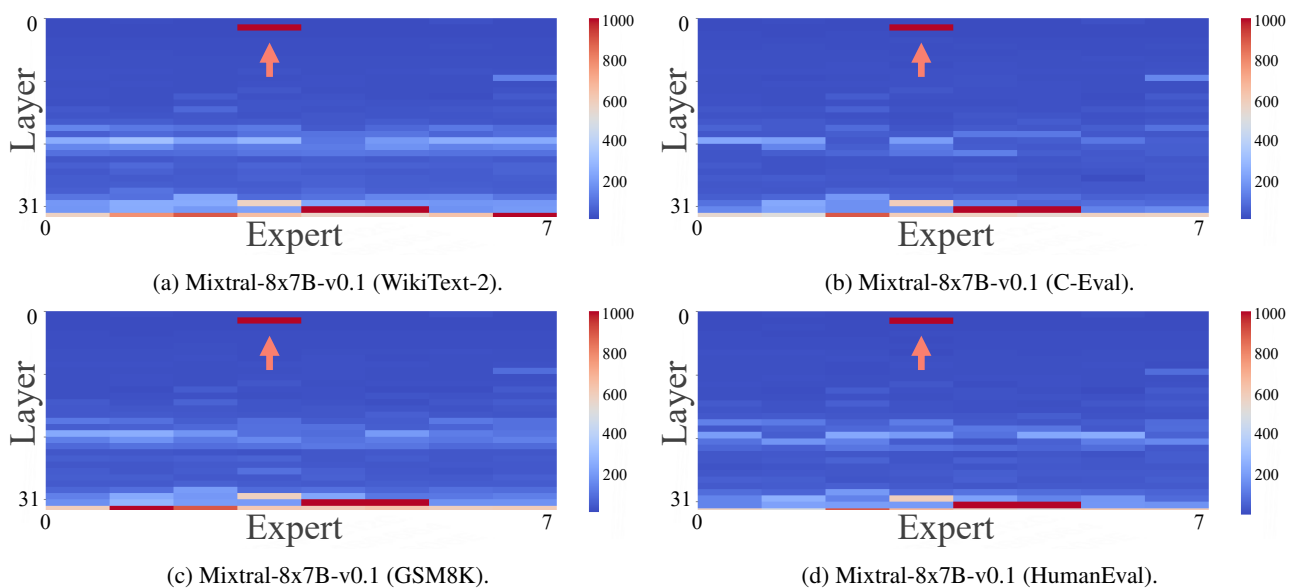


Figure 13: Heatmap visualizations of the maximum output magnitudes from the `down_proj` for each expert in Mixtral-8x7B-v0.1 across multiple datasets. SEs are highlighted with arrows.